



11320 - Constraining the Star Forming Main Sequence Relation in the Early Universe through Archival Pure Parallel Observations

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

A major goal of extragalactic astrophysics is to piece together our understanding of star formation through cosmic time. Our current understanding of star formation at high redshifts has been poorly constrained by prior observational limitations; however, with the unprecedented sensitivity and resolution of JWST, we are now uncovering galaxies that challenge long-standing theoretical models of how the first generations of stars and galaxies formed and evolved. We propose to constrain the low mass end of the star forming main sequence (SFMS) at $z > 5$ in order to piece together our understanding of galaxy formation through cosmic time. We will utilize stellar population synthesis modeling codes to simultaneously infer stellar properties and model star formation histories by fitting both the photometric and spectroscopic data from NIRCам WFSS pure-parallel observations. Because these observations of galaxies cover many different areas of the sky and are untargeted, we are able to probe an unbiased range of galaxies in the early Universe. We will also investigate the 'bursty' star formation in low-mass, high redshift galaxies through a comparison of star

formation rates derived from line fluxes and SED fitting, in order to assess how bursty star formation contributes to the intrinsic scatter of the SFMS and give us insight into early Universe star formation.

OBSERVING DESCRIPTION